

Password Security & Cracking Lab Report

1. Introduction

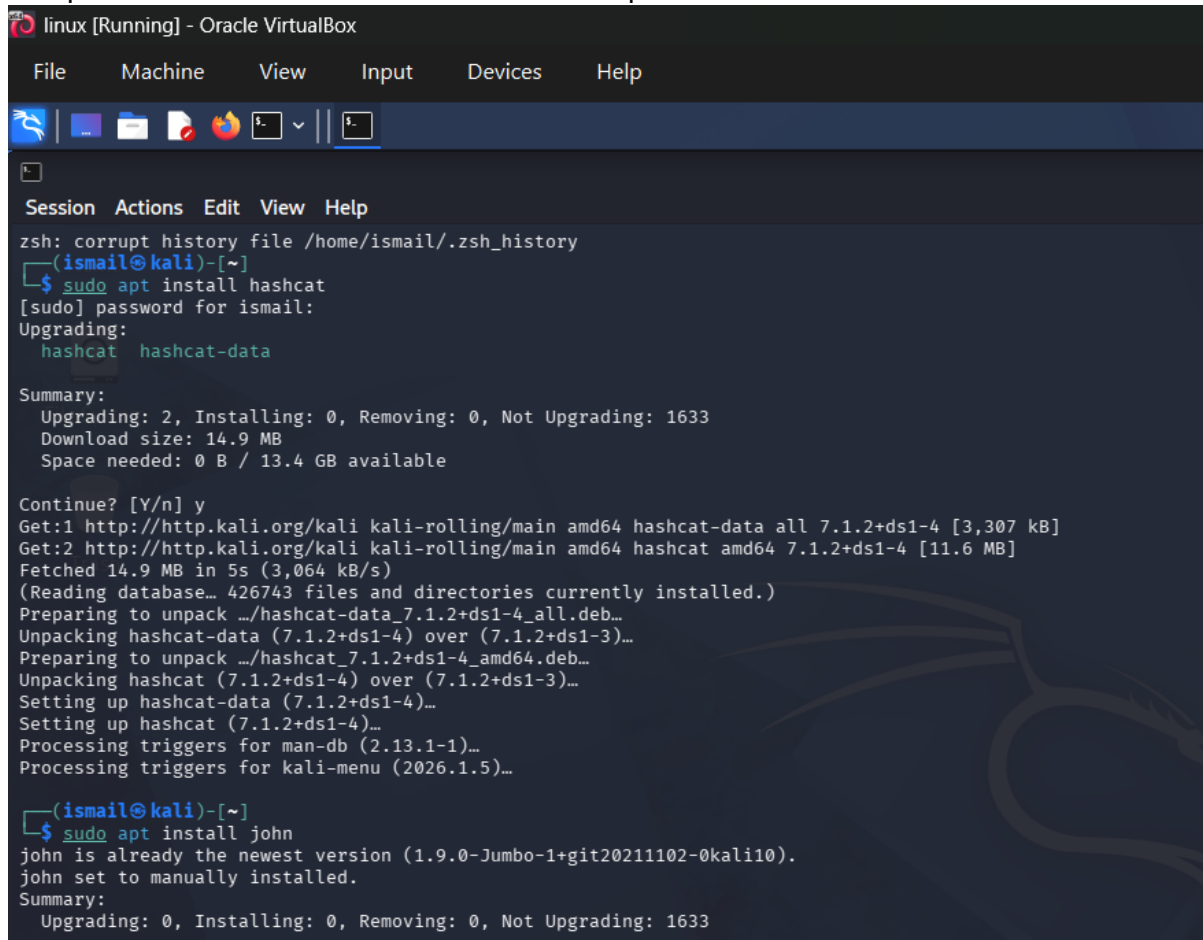
This lab demonstrates password hashing and cracking using Kali Linux tools.

Tools used: John the Ripper, Hashcat, OpenSSL.

Environment: Kali Linux Virtual Machine.

2. Methodology

Step 1: Tool Installation and Environment Setup



```
linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Session Actions Edit View Help
zsh: corrupt history file /home/ismail/.zsh_history
(ismail@kali)~]
$ sudo apt install hashcat
[sudo] password for ismail:
Upgrading:
  hashcat  hashcat-data
Summary:
  Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 1633
  Download size: 14.9 MB
  Space needed: 0 B / 13.4 GB available
Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 hashcat-data all 7.1.2+ds1-4 [3,307 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 hashcat amd64 7.1.2+ds1-4 [11.6 MB]
Fetched 14.9 MB in 5s (3,064 kB/s)
(Reading database... 426743 files and directories currently installed.)
Preparing to unpack .../hashcat-data_7.1.2+ds1-4_all.deb...
Unpacking hashcat-data (7.1.2+ds1-4) over (7.1.2+ds1-3)...
Preparing to unpack .../hashcat_7.1.2+ds1-4_amd64.deb...
Unpacking hashcat (7.1.2+ds1-4) over (7.1.2+ds1-3)...
Setting up hashcat-data (7.1.2+ds1-4)...
Setting up hashcat (7.1.2+ds1-4)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...
(ismail@kali)~]
$ sudo apt install john
john is already the newest version (1.9.0-Jumbo-1+git20211102-0kali10).
john set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1633
```

Figure 1: Installation of Hashcat and John the Ripper

Step 2: Wordlist Verification (rockyou.txt)

```
linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Minimize all open windows and show the desktop
Session Actions Edit View Help

ismail@kali: ~
└─$ ls /usr/share/wordlists/
dirb      dnsmap.txt  fern-wifi  legion     nmap.lst   sqlmap.txt  wifite.txt
dirbuster fasttrack.txt john.lst   metasploit rockyou.txt.gz wfuzz

(ismail@kali)-[~]
└─$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz
[sudo] password for ismail:

(ismail@kali)-[~]
└─$ john --wordlists=/usr/share/wordlists/rockyou.txt weak.txt
Created directory: /home/ismail/.john
Unknown option: "--wordlists=/usr/share/wordlists/rockyou.txt"

(ismail@kali)-[~]
└─$ john --show weak.txt
stat: weak.txt: No such file or directory

(ismail@kali)-[~]
└─$ ls
Desktop  Downloads  lynis.pid  Music      pass.txt  Pictures  Templates  Videos
Documents  lynis.log  lynis-report.dat  output.gnmap  passwd.txt  Public    test.txt

(ismail@kali)-[~]
└─$ echo -n "123" | sha256sum | awk '{print $1}' > weak.txt

(ismail@kali)-[~]
└─$ echo -n "mypassword123" | sha256sum | awk '{print $1}' > medium.txt

(ismail@kali)-[~]
└─$ echo -n "Myp@ssword#122" | sha256sum | awk '{print $1}' > strong.txt

(ismail@kali)-[~]
└─$ ls
Desktop  Downloads  lynis.pid  medium.txt  output.gnmap  passwd.txt  Public  Templates  Videos
Documents  lynis.log  lynis-report.dat  Music      pass.txt     Pictures    strong.txt  test.txt  weak.txt

(ismail@kali)-[~]
└─$ cat weak.txt
a665a45920422f9d417e4867efdc4fb8a04a1f3fff1fa07e998e86f7f7a27ae3

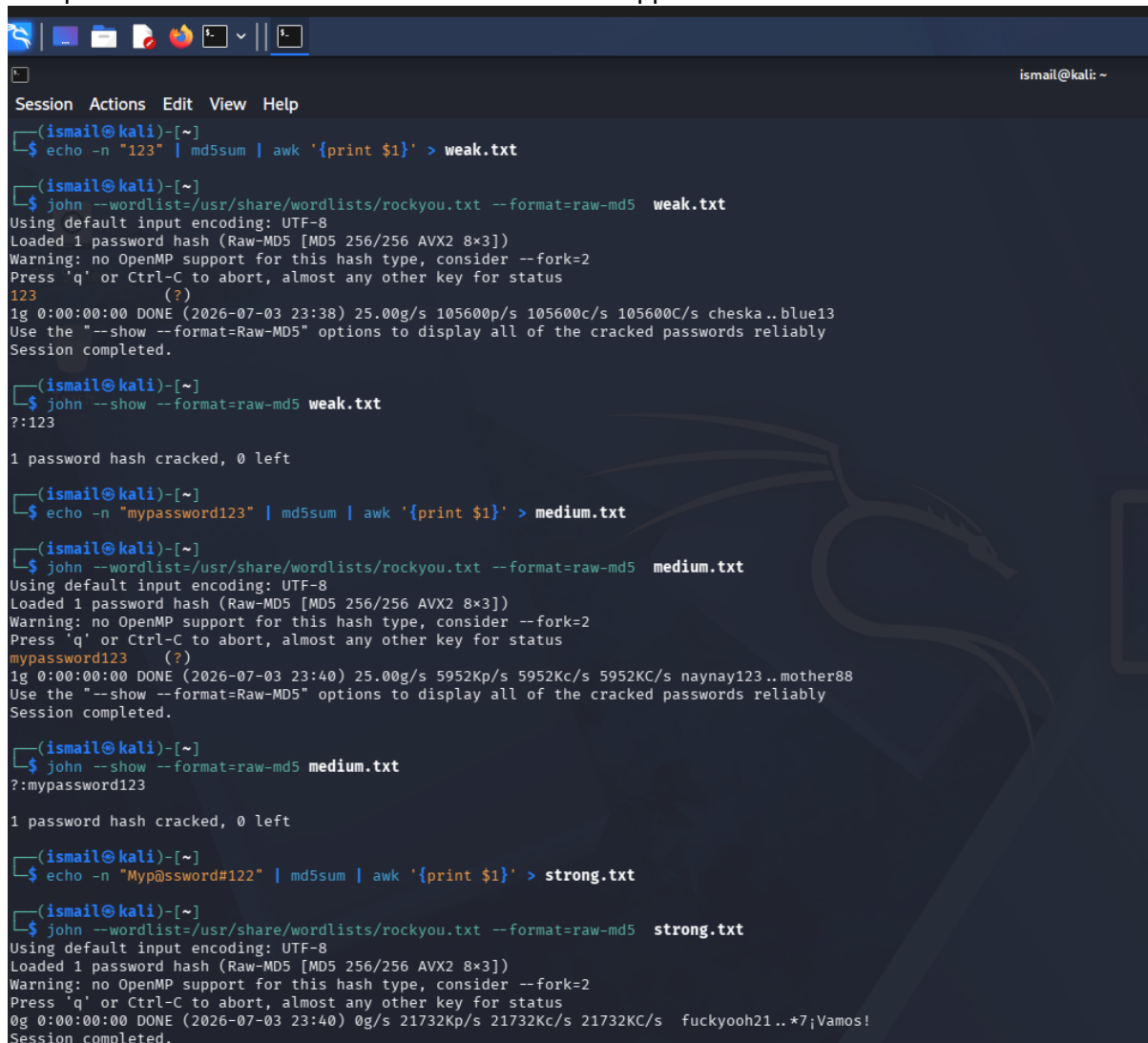
(ismail@kali)-[~]
└─$ cat medium.txt
6e659deaa85842cdabb5c6305fcc40033ba43772ec00d45c2a3c921741a5e377

(ismail@kali)-[~]
└─$ cat strong.txt
912ae298d1af5da180e27864e079d3da52760eb791dfb11dc34b443d5c7c2ad6

(ismail@kali)-[~]
└─$ john --wordlist=/usr/share/wordlists/rockyou.txt weak.txt
Warning: detected hash type "cryptoSafe", but the string is also recognized as "gost"
```

Figure 2: RockYou wordlist extraction and verification

Step 3: Password Hash Generation & John the Ripper

A terminal window titled 'ismail@kali: ~' showing a series of commands and outputs for password cracking. The user generates three files: 'weak.txt' (hash of '123'), 'medium.txt' (hash of 'mypassword123'), and 'strong.txt' (hash of 'Myp@ssword#122'). For each file, they run 'john --wordlist=/usr/share/wordlists/rockyou.txt --format=raw-md5' and then 'john --show --format=raw-md5'. The output for 'weak.txt' shows the password '123' cracked. The output for 'medium.txt' shows the password 'mypassword123' cracked. The output for 'strong.txt' shows the password 'fuckyooh21..*7;Vamos!' cracked. The terminal has a dark background with a faint Kali Linux dragon logo.

```
(ismail@kali)-[~]
$ echo -n "123" | md5sum | awk '{print $1}' > weak.txt

(ismail@kali)-[~]
$ john --wordlist=/usr/share/wordlists/rockyou.txt --format=raw-md5 weak.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Press 'q' or Ctrl-C to abort, almost any other key for status
123 (?)
1g 0:00:00:00 DONE (2026-07-03 23:38) 25.00g/s 105600p/s 105600c/s 105600C/s cheska..blue13
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords reliably
Session completed.

(ismail@kali)-[~]
$ john --show --format=raw-md5 weak.txt
?:123

1 password hash cracked, 0 left

(ismail@kali)-[~]
$ echo -n "mypassword123" | md5sum | awk '{print $1}' > medium.txt

(ismail@kali)-[~]
$ john --wordlist=/usr/share/wordlists/rockyou.txt --format=raw-md5 medium.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Press 'q' or Ctrl-C to abort, almost any other key for status
mypassword123 (?)
1g 0:00:00:00 DONE (2026-07-03 23:40) 25.00g/s 5952Kp/s 5952Kc/s 5952KC/s naynay123..mother88
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords reliably
Session completed.

(ismail@kali)-[~]
$ john --show --format=raw-md5 medium.txt
?:mypassword123

1 password hash cracked, 0 left

(ismail@kali)-[~]
$ echo -n "Myp@ssword#122" | md5sum | awk '{print $1}' > strong.txt

(ismail@kali)-[~]
$ john --wordlist=/usr/share/wordlists/rockyou.txt --format=raw-md5 strong.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:00 DONE (2026-07-03 23:40) 0g/s 21732Kp/s 21732Kc/s 21732KC/s fuckyooh21..*7;Vamos!
Session completed.
```

Figure 3: Password cracking using John the Ripper

Step 4: Hashcat Cracking (Weak & Medium)

```
Session Actions Edit View Help

--(ismail@kali)-[~]
--$ hashcat -m 0 weak.txt /usr/share/wordlists/rockyou.txt --force
hashcat (v7.1.2) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_D
- Platform #1 [The pocl project]

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Device #01: cpu-haswell-11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz, 739/1478 MB (256 MB allocatable), 2MB
Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
  Zero-Byte
  Early-Skip
  Not-Salted
  Not-Iterated
  Single-Hash
  Single-Salt
  Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Host memory allocated for this attack: 512 MB (805 MB free)

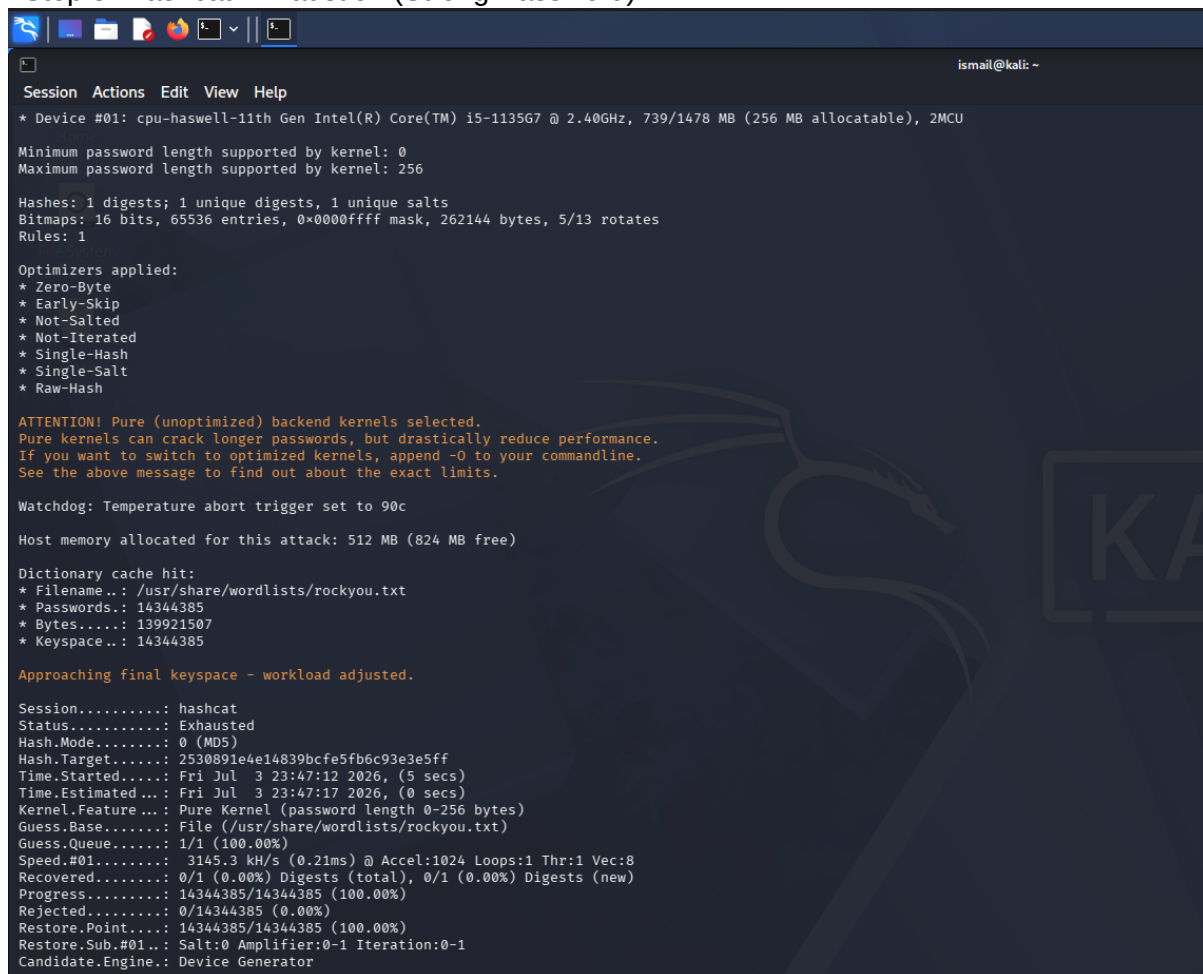
Dictionary cache built:
  Filename..: /usr/share/wordlists/rockyou.txt
  Passwords.: 14344392
  Bytes.....: 139921507
  Keyspace..: 14344385
  Runtime ...: 1 sec

02cb962ac59075b964b07152d234b70:123

Session.....: hashcat
Status.....: Cracked
```

Figure 4: Hashcat cracking weak and medium passwords

Step 5: Hashcat Exhaustion (Strong Password)



The screenshot shows a Kali Linux terminal window with the title bar 'ismail@kali: ~'. The terminal displays the output of a Hashcat command. The output includes system information, password length limits, hash details, and a list of optimizers. A warning message states: 'ATTENTION! Pure (unoptimized) backend kernels selected. Pure kernels can crack longer passwords, but drastically reduce performance. If you want to switch to optimized kernels, append -O to your commandline. See the above message to find out about the exact limits.' The terminal also shows the watchdog temperature, host memory allocation, and dictionary cache hit information. The final status is 'Exhausted'.

```
Session Actions Edit View Help
* Device #01: cpu-haswell-11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz, 739/1478 MB (256 MB allocatable), 2MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Early-Skip
* Not-Salted
* Not-Iterated
* Single-Hash
* Single-Salt
* Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

Host memory allocated for this attack: 512 MB (824 MB free)

Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 14344385

Approaching final keyspace - workload adjusted.

Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 0 (MD5)
Hash.Target.....: 2530891e4e14839bcfe5fb6c93e3e5ff
Time.Started.....: Fri Jul 3 23:47:12 2026, (5 secs)
Time.Estimated...: Fri Jul 3 23:47:17 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 3145.3 kH/s (0.21ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 14344385/14344385 (100.00%)
Rejected.....: 0/14344385 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
```

Figure 5: Strong password resisted cracking attempt

3. Observations

Weak passwords are cracked easily. Medium passwords show moderate resistance. Strong passwords resist cracking attempts.

MD5 is insecure and should not be used in production systems.

4. Recommendations

Use strong passwords with complexity and length.

Use modern hashing algorithms like bcrypt or Argon2.

Enable multi-factor authentication for better security.

5. Conclusion

This lab demonstrates the importance of password strength and secure hashing algorithms in cybersecurity.